Operations Research

The traveling salesman problem

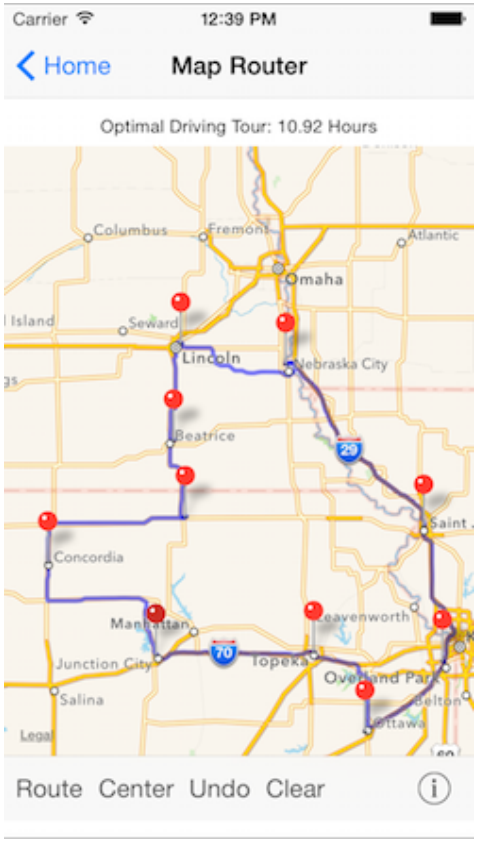
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Traveling salesman

Input. Customers to visit:



Task. Find the shortest tour visiting all customers.

Scientific achievements

Performance evolution for the traveling salesman problem:

1954	Dantzig, Fulkerson, Johnson	**49** cities
1971	Held, Karp	**64** cities
1975	Camerini, Fratta, Maffioli	**67** cities
1977	Grötschel	**120** cities
1980	Crowder, Padberg	**318** cities
1987	Padberg, Rinaldi	**532** cities
1987	Grötschel, Holland	**666** cities
1987	Padberg, Rinaldi	**2,392** cities
1994	Applegate, Bixby, Chvátal, Cook	**7,397** cities
1998	Applegate, Bixby, Chvátal, Cook	**13,509** cities
2001	Applegate, Bixby, Chvátal, Cook	**15,112** cities
2004	Applegate, Bixby, Chvátal, Cook, Helsgaun	**24,978** cities
2016	Cook, Espinoza, Goycoolea, Helsgaun,	**49,603** cities

And the main reason of this progress is **not** computer improvement.

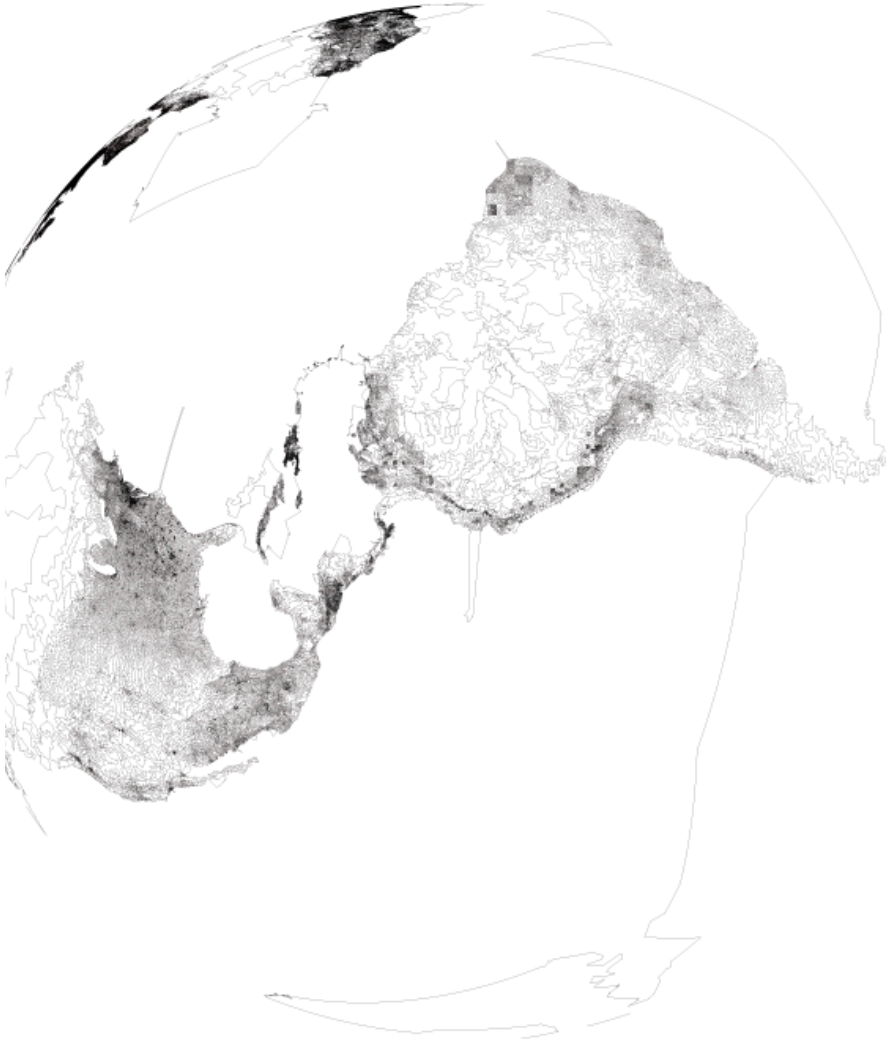
Combinatorial explosion

Solving the TSP on n cities: number of possible tours is $(n-1)!$.

With a computer testing 10^{10} tours per second:

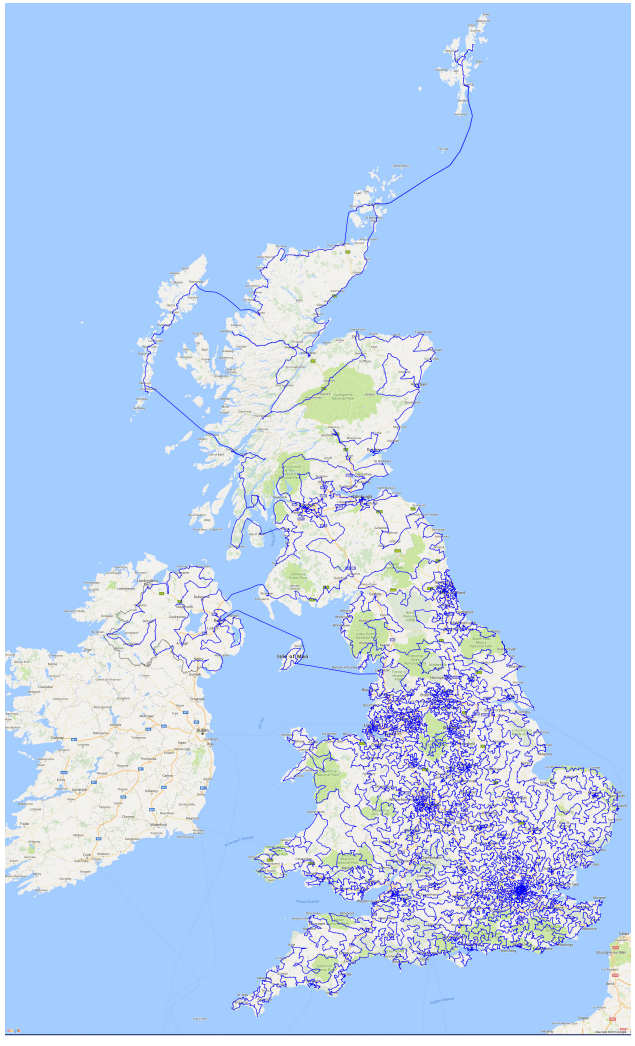
Number of cities	Number of tours	Solving time
6	**120**	**12 ns**
10	**362,880**	**36 μs**
20	1.22×10^{17}	**141** days
50	6.08×10^{62}	1.93×10^{45} years
100	9.33×10^{155}	2.96×10^{96} years

Visiting all cities in the world



1,904,711 cities; proposed solution: **7,516,353,779 km**; proved to be within **0.076%** of the optimal solution in 2003!

Visiting **24,727** pubs in the United Kingdom

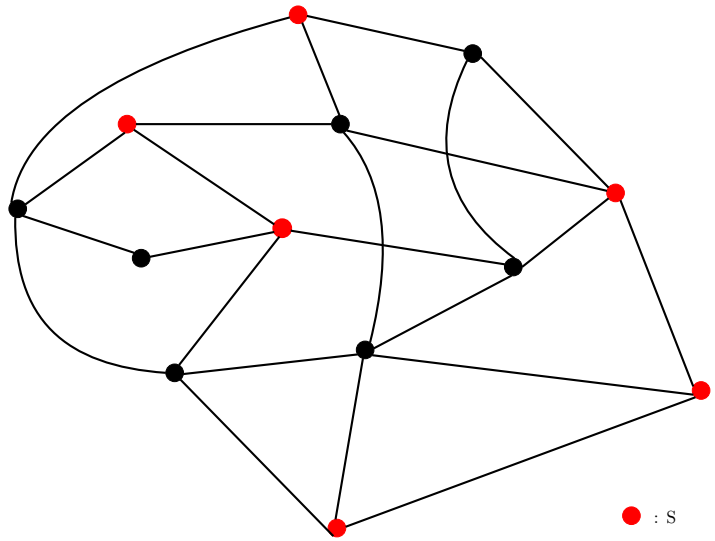


Optimal solution: **45,495,239 m** !

First modeling

Input. Graph $G = (V, E)$ endowed with a "length" $\ell : E \to \mathbb{R}_+$, a subset $S \subseteq V$ of vertices (to visit).

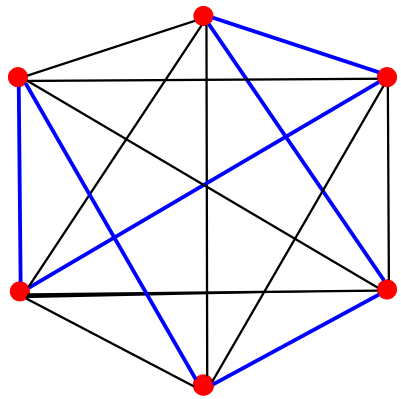
Task. Find the walk C of G, visiting all vertices in S, of minimal length.



Better modeling

Input. Graph $H = (S, F)$, endowed with a "distance" $d : F \to \mathbb{R}_+$.

Task. Find the hamiltonian cycle C of H with minimal total distance.



If we require that d satisfy triangular inequality, the two modelings are equivalent:

$$d(uv) := \text{length of the shortest } u\text{-}v \text{ path in } G.$$

Two IP modelings

$$\textbf{Min} \quad \sum_{u,v \in S:\ u \neq v} d(uv) z_{u,v}$$

s.c.

$$\sum_{u \neq v} z_{u,v} = 1 \qquad \forall v \in S$$

$$\sum_{u \neq v} z_{v,u} = 1 \qquad \forall v \in S$$

$$y_u + (1+n) z_{u,v} \leqslant y_v + n \qquad \forall u, v \in S,\ u \neq v,\ v \neq o$$

$$z_{u,v} \in \{0,1\} \qquad \forall u, v \in S, u \neq v$$

$$y_v \in \mathbb{R}_+ \qquad \forall v \in S$$

Can be directly solved with a free or commercial solver.

Can be adapted to other situations (with a time delay on the edges for instance).

Cannot solve large instances (very bad linear relaxation).

$$\textbf{Min} \quad \sum_{e \in F} d(e) x_e$$

s.c.

$$\sum_{e \in \delta(v)} x_e = 2 \qquad \forall v \in S$$

$$\sum_{e \in \delta(X)} x_e \geqslant 2 \qquad \forall X \subseteq S \text{ tq } X \notin \{\varnothing, S\}$$

$$x_e \in \{0,1\} \qquad \forall e \in F$$

Cannot be directly solved with a free or commercial solver.

Does not adapt easily to other situations.

Can solve huge instances (good linear relaxation).

Efficient formulation

$$\textbf{Min} \quad \sum_{e \in F} d(e) x_e$$

$$\text{s.c.} \quad \sum_{e \in \delta(v)} x_e = 2 \quad \forall v \in S$$

$$\sum_{e \in \delta(X)} x_e \geqslant 2 \quad \forall X \subseteq S \text{ tq } X \notin \{\varnothing, S\}$$

$$x_e \in \{0, 1\} \quad \forall e \in F$$

Constraint separation

Main idea: we suppose given an algorithm $\mathcal{A}$ that solves

CONSTRAINT SEPARATION FOR TSP

Input. Vector $x \in \mathbb{R}^E$.

Task. Return 'yes' if x satisfies all the constraints $\sum_{e \in \delta(X)} x_e \geqslant 2$ for $X \in \mathcal{C}$. If not, return $X' \in \mathcal{C}$ such that $\sum_{e \in \delta(X')} x_e < 2$.

Branch-and-cut

We perform a branch-and-bound. We choose $\mathcal{C}' \subseteq \mathcal{C}$.

In each node of the branch-and-bound tree:

1. We solve

$$\begin{array}{lll} \textbf{Min} & \displaystyle\sum_{e \in E} c(e) x_e & \\ \text{s.c.} & 0 \leqslant x_e \leqslant 1 & \forall e \in E, \\ & \displaystyle\sum_{e \in \delta(v)} x_e = 2 & \forall v \in V, \\ & \displaystyle\sum_{e \in \delta(X)} x_e \geqslant 2 & \forall X \in \mathcal{C}'. \end{array}$$

   This provides a solution $\bar{x}$.
2. We apply $\mathcal{A}$ on $\bar{x}$.
   - If 'yes', then we resume the branch-and-bound (*the linear relaxation of* (P) *is solved*).
   - Otherwise, we go back to 1. and add the X' obtained by $\mathcal{A}$ to $\mathcal{C}'$.

Constraint separation for TSP

There is an algorithm $\mathcal{A}$ separating the TSP constraints:

Put $\bar{x}_e$ as a capacity on e, for all $e \in E$.

Fix s in V; for each t in $V \setminus \{s\}$, compute an s-t cut of minimal capacity.

If none of the cuts $\delta(X')$ has capacity < 2, then return 'yes'. Else, return 'no' and X'.